1D CNN — Architecture Variant Comparison

Generated 2026-05-15 11:51:48

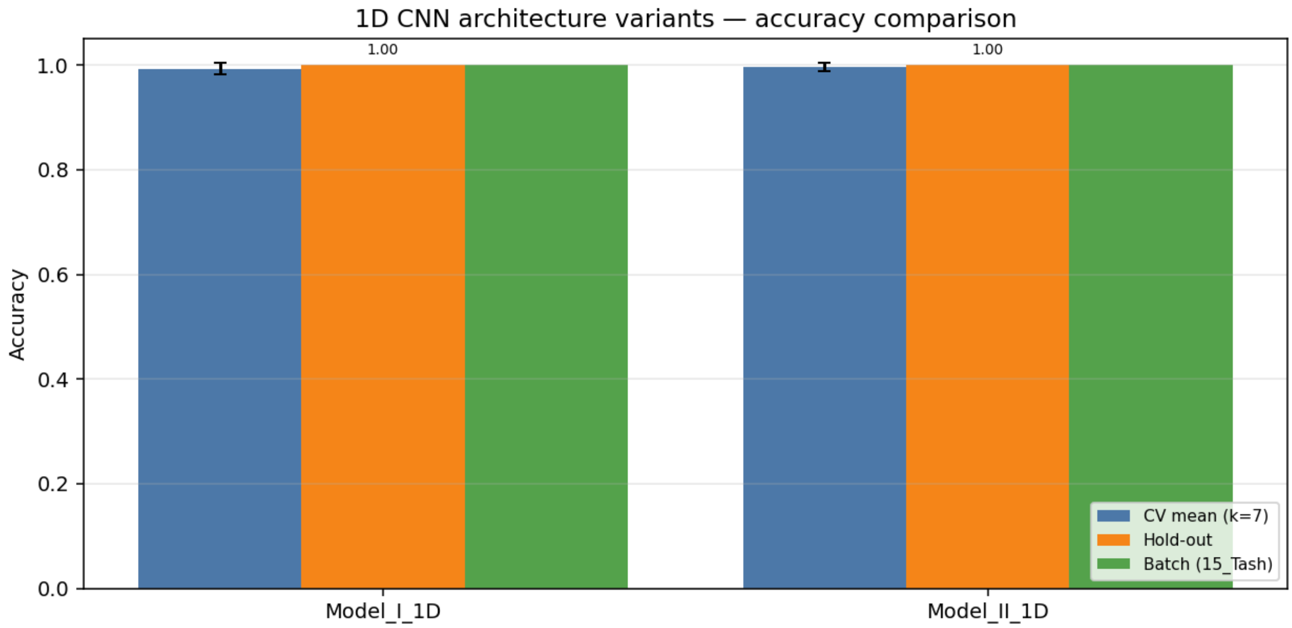
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	72 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

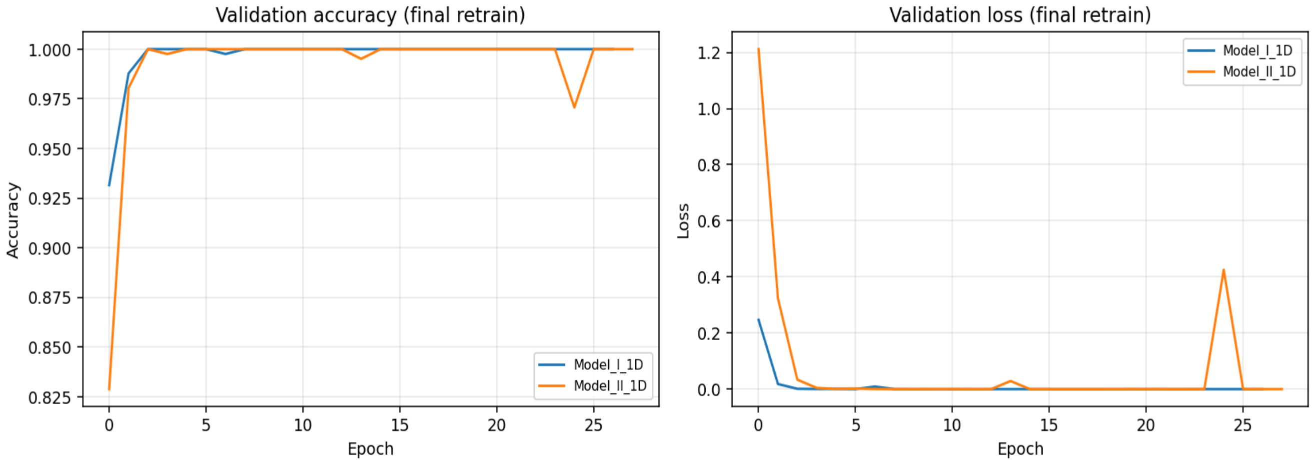
Variant comparison (summary)

Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	126,950	0.9932	0.0108	1.0000	0.0000	1.0000	30/30
Model_II_1D	82,246	0.9966	0.0083	1.0000	0.0000	1.0000	30/30

Best on hold-out: **Model_I_1D** · Best on CV: **Model_II_1D**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Model_I_1D
Model_II_1D
Validation loss (final retrain)
Loss
Epoch
Model_I_1D
Model_II_1D

Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	126,950
CV mean ± std	0.9932 ± 0.0108
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	1.0000 (30/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	1.0000	1.0000	1.0000
2	1750	42	1.0000	1.0000	1.0000
3	1750	42	0.9762	0.9762	0.9762
4	1750	42	1.0000	1.0000	1.0000
5	1750	42	0.9762	0.9762	0.9524
6	1757	41	1.0000	1.0000	1.0000
7	1757	41	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	**30**	**30**	**1.000**

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM): Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	82,246
CV mean ± std	0.9966 ± 0.0083
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	1.0000 (30/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	1.0000	1.0000	1.0000
2	1750	42	1.0000	1.0000	1.0000
3	1750	42	0.9762	0.9762	0.9762
4	1750	42	1.0000	1.0000	1.0000
5	1750	42	1.0000	1.0000	0.9762
6	1757	41	1.0000	1.0000	0.9756
7	1757	41	1.0000	1.0000	0.9756

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	**30**	**30**	**1.000**